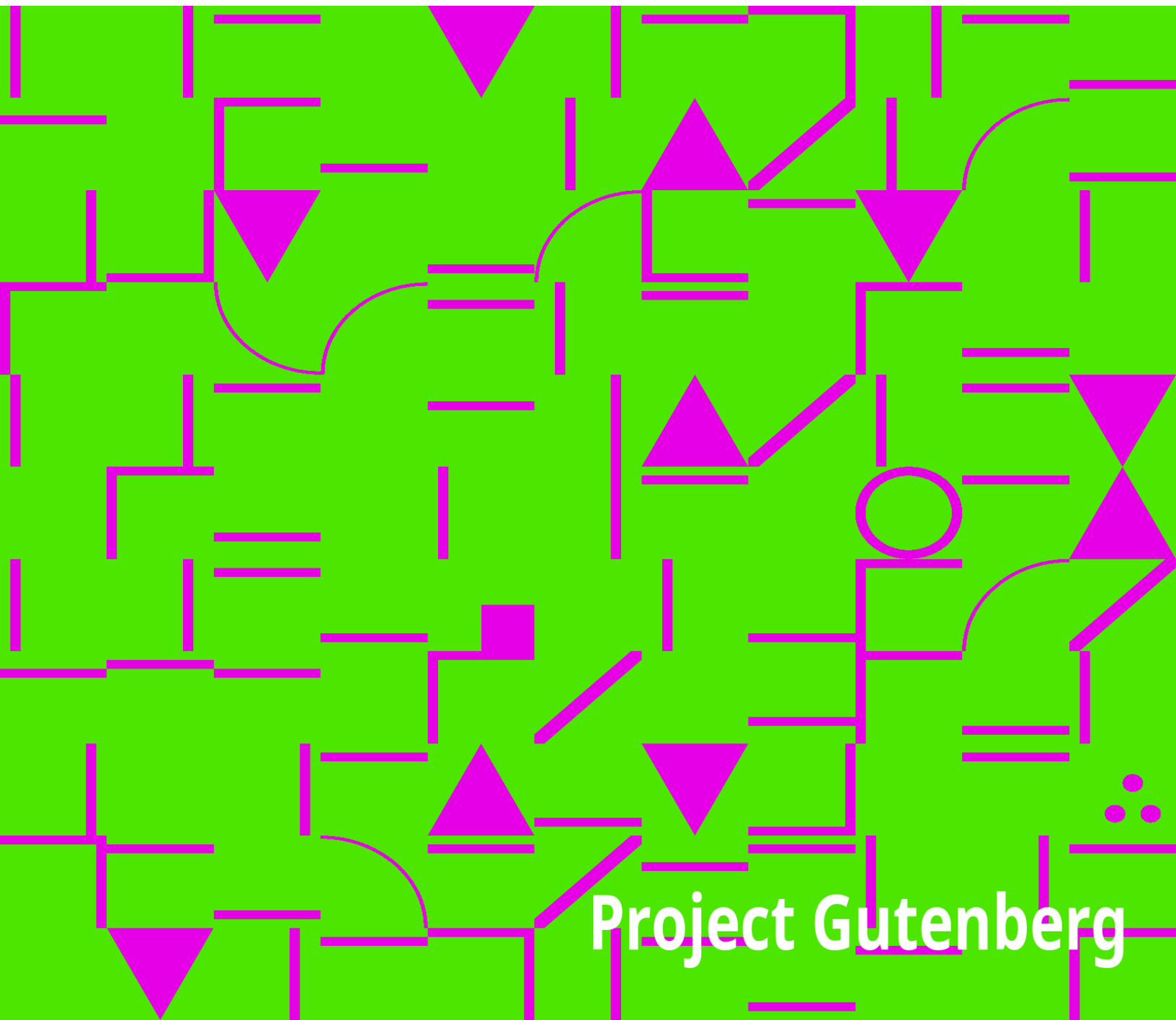


The Shepherd of Banbury's Rules to Judge of the Changes of the Weather, Grounded on Forty Years' Experience

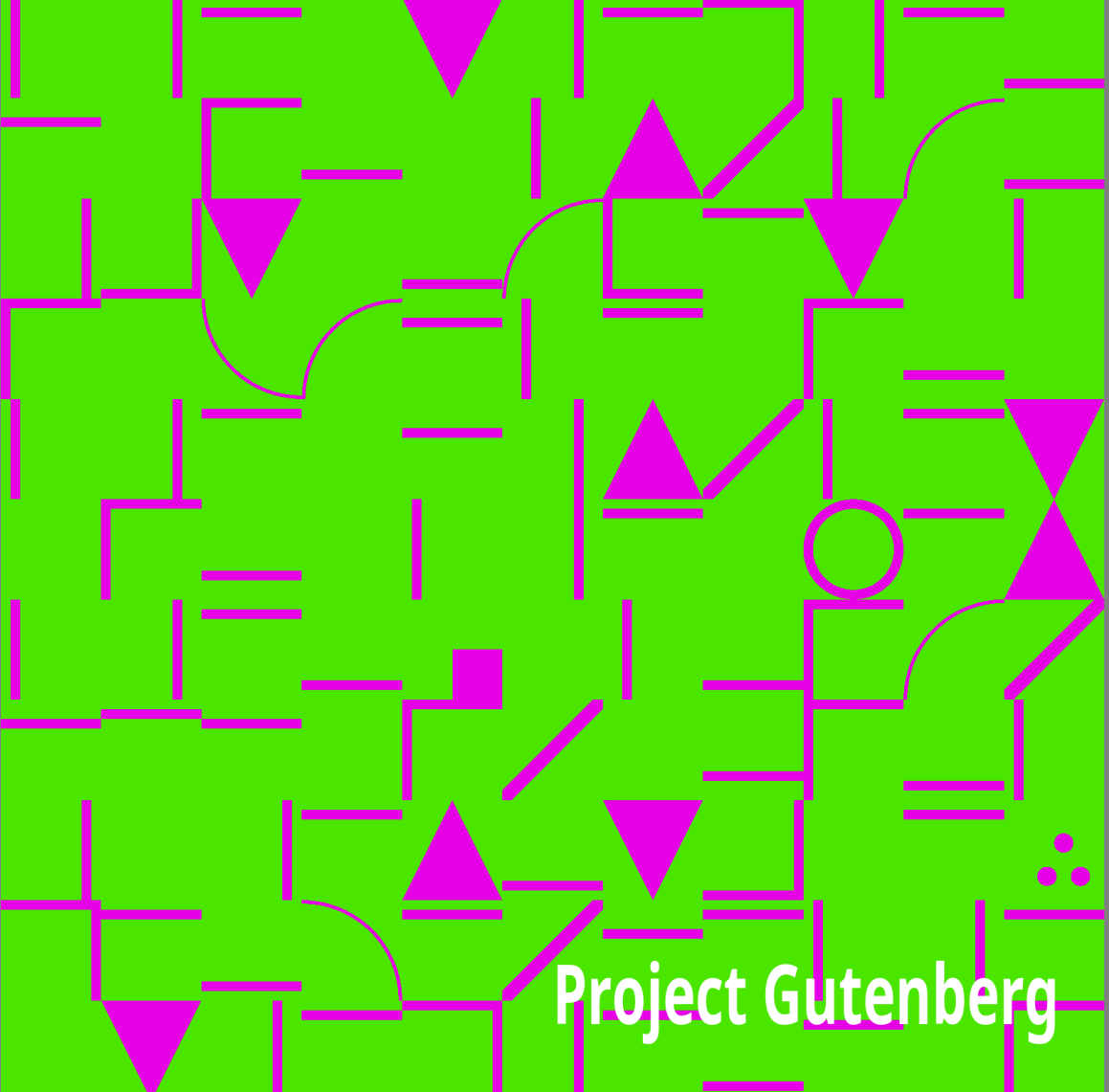
John Claridge

An abstract geometric pattern composed of various shapes including triangles, squares, circles, and lines in shades of blue, green, and yellow, set against a black background.

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*** START OF THE PROJECT GUTENBERG EBOOK THE SHEPHERD
OF BANBURY'S RULES TO JUDGE OF THE CHANGES OF THE
WEATHER, GROUNDED ON FORTY YEARS' EXPERIENCE ***

THE
SHEPHERD
OF

BANBURY'S RULES

To judge of the
CHANGES of the WEATHER,
Grounded on Forty Years' EXPERIENCE;
By which you may know
The WEATHER for several Days to come,
and in some Cases for Months.

To which is added
A Rational ACCOUNT of the CAUSES of
such Alterations, the Nature of Wind,
Rain, Snow, &c.

By *JOHN CLARIDGE*, Shepherd.

A NEW EDITION, Corrected.

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INTRODUCTION.

AS we very justly esteem it a fit Tribute of Admiration to adorn natural Curiosities, by setting them as richly and as advantageously as art can direct, so the following Observations of the Shepherd of *Banbury* have appeared to me worthy of being presented to the Eye of the Public, with all the Lustre that it was in my Power to give them. It is one thing to observe, and another to reason upon Observations, and it very rarely happens that both can be taken into the Compass of one Man's Life. We ought therefore to consider it as a very lucky Incident, when the Observations of another Man, upon whom we can depend, fall into our Hands, and enable us to add natural Experience to the Notions derived to us from Books of Philosophy.

THERE is a Degree of Pedantry in Desarts as well as Colleges. Men who derive their Knowledge entirely from Experience are apt to despise what they call Book Learning, and Men of great Reading are as apt to fall into a less excusable mistake, that of taking the Knowledge of Words for the Knowledge of Things; whereas there are not any two points more opposite in Nature, since we very rarely see, that either true Scholars are talkative, or that talkative Men are true Scholars.

THE Shepherd, whose sole Business it is to observe what has a Reference to the Flock under his Care, who spends all his Days and many of his Nights in the open Air, and under the wide spread Canopy of Heaven, is in a Manner obliged to take particular Notice of the Alterations of the Weather, and when once he comes to take a Pleasure in making such Observations, it is amazing how great a Progress he makes in them, and to how great a

Certainty at last he arrives by mere dint of comparing Signs and Events, and correcting one Remark by another. Every thing in Time becomes to him a Sort of Weather-Gage. The Sun, the Moon, the Stars, the Clouds, the Winds, the Mists, the Trees, the Flowers, the Herbs, and almost every Animal with which he is acquainted. All these I say become to such a person Instruments of real Knowledge.

THERE are a Sort of half wise People, who from the Consideration of the Distances of Things, are apt to treat such Prognostications, as they phrase them, with much Contempt. They can see no Connexion between a Cat's washing her Face, and the Sky's being overspread with Clouds, and therefore they boldly pronounce that the one has no Relation to the other. Yet the same People will readily own that the fluttering of the Flame of a Candle is a certain token of Wind, which however is not discernible by their Feeling; because it lies within the Compass of their Understanding to discern that this Fluctuation of the Flame is caused by the Wind acting upon it, and therefore they are inclined to believe this, though it does not fall actually under the Cognizance of their Senses. But a Man of a larger Compass of Knowledge, who is acquainted with the Nature and Qualities of the Air, and knows what an Effect any Alterations in the Weight, the Dryness, or the Humidity of it has upon all animal Bodies, easily perceives the Reason why other Animals are much sooner sensible of any Alterations that happen in that Element than Men, and therefore to him the cawing of Ravens, the chattering of Swallows, and a Cat's washing her Face are not superstitious Signs, but natural tokens (like that of the Candle's fluttering) of a Change of Weather, and as such they have been thought worthy of Notice by *Aristotle*, *Virgil*, *Pliny*, and all the wisest and gravest Writers of Antiquity.

BUT still a few slight and trivial Observations of this Kind, and such as are in the Power of every Man to make, go but a very little Way in furnishing us with a useful Knowledge of the Indications of the Weather. To supply these, and to have constantly at Hand the Means of judging of these Alterations,

Men of great Genius have invented, and wonderful Inventions they are! Instruments for measuring the Heat, the Cold, the Weight, the Dryness, and the Humidity of the Air, with great Exactness, and upon these they reason as to the changes of Weather with great Accuracy and Certainty. It would undoubtedly be a great Folly to pretend to question either the Truth of their Observations, or the Usefulness of them: but then we may have leave to consider how far, and to how great a Degree they are useful. The Thermometer measures exactly the Degrees of Heat, but the Air must be hot to such or such a Degree before it is discerned by this Instrument. The barometer indicates the Weight of the Air, and the rising and falling of the Quicksilver expresses the Alterations in its Weight with wonderful Nicety, but then those Alterations are the Cause of this. In like manner the Hygrometer, or Hygroscope, measures the Dryness or the Humidity of the Air very plainly and very exactly, but the Weather must alter, must become dryer or moister than it was, before these Alterations are visible; and therefore, however ingenious, however curious, however useful these Instruments may be in other Respects, they undoubtedly contribute very little to the prognosticating a Change of Weather at a Distance; and it is from the Experience of this, that they are so little esteemed, so lightly regarded by the common People.

OUR Shepherd's Observations are of quite another Nature, most of them give us a Day's Notice, many a Week's, and some extend to several Months' Prognostication of the Changes of the Weather, and of how great Use these may be to all Ranks and Degrees of People, to the sedentary Valetudinarian, as well as the active Traveller, to the Sportsman who pursues his Game, as well as to the industrious Husbandman who constantly follows his Labour; in short, to every Man in every Situation in some Degree or other, is so very clear and intelligible, that it would, be a mere waste of Words, and a very idle display of Rhetoric, to attempt the making it clearer. Every Man living would be glad to foresee the Alterations of Weather if he could, and consequently to most People, if not to all, these Observations, grounded on no less than forty Years' Experience, cannot but be acceptable.

TO make the best use of one's Talent, and to employ the Lights derived from the Station in which Providence has placed one for the Benefit of Mankind, is undoubtedly discharging one's Duty, answering the End of our Creation, and corresponding with the Œconomy of Nature, which does nothing in vain. This Proposition is equally true, let a Man's Station be what it will. It is the Manner in which we perform, and not the Character, that makes the Player, and in this Sense what Man is not a Player? Here then is an Instance of one who has for many Years studied his Part, and now communicates his Discoveries freely. In a Physician, in a Philosopher, in a Mathematician, this would be highly commendable, and why not in a Shepherd? We do not cast our own Parts in the Drama of Life; no, this is performed by the great Author of Nature. He who adjusted every Thing on Earth with such Beauty and Harmony, he who taught the Heavenly Bodies to move; the same distributed their several Offices to Men. May we not therefore suppose that every Man's Part is well cast, and that our Abilities are exactly proportioned to our Stations? If so, he who does all he can, does all that ought to be expected from him, and merits from impartial Judges the most general and just Applause. To be convinced of this, we need not only reflect on the narrow and selfish Conduct of some, who either by Study or by Chance, have acquired certain valuable Secrets, which with the utmost Industry they conceal in order to be the more admired, or that they may render them beneficial to themselves. How contrary the Conduct of our Shepherd! His Pains were all his own, but the Fruit of them he thus generously offers to the Public. Good Sense and the dictates of Nature taught him this Maxim, *That what might benefit many, should not be concealed by one from Views of Profit or of Pride.*

IN my Remarks upon the Shepherd's Rules, I have sometimes endeavoured to support them by Authorities, which I must confess would have been of little Use if the Author had been a Person of Learning; but when it is considered that these Observations were purely the Effect of his own Attention and Experience, it certainly strengthens them, and adds greatly to their Credit that they have been esteemed evident Signs of the same Effects,

by the greatest Masters in this Kind of Science. The Art of prognosticating the Weather may be considered as a Kind of decyphering, and in that Art it is always allowed a point of great Consequence, when several Masters therein agree as to the meaning of a Character, and it is from thence very justly presumed that this Character is rightly decyphered.

I have also endeavoured to explain most of his Observations, according to the Rules of the new Philosophy, which, as it is grounded upon Experiments, so it generally speaking enables us to give a fair and rational Account of almost all the Phænomena taken notice of by the Shepherd of *Banbury*.

I likewise have added some other Rules in Relation to the Weather, taken from the common sayings of our Country People, and from old *English* Books of Husbandry, but I have distinguished all these from the Observations themselves, so that the Reader will have no Trouble to discern the Text from the Commentary, or to know what belongs to the Shepherd of *Banbury*, and what to the Editor of his Observations. This I think may serve by the Way of Introduction, let us now proceed to the Rules themselves.

THE

Country Calendar,
OR THE
SHEPHERD OF BANBURY'S
OBSERVATIONS.

I.

SUN. *If the Sun rise red and firey.* } Wind and Rain.

THE Reason of this Appearance is, because the Sun shines through a large Mass of Vapours, which occasions that red Colour that has been always esteemed a Sign of Rain, especially if the Face of the Sun appear bigger than it ought, for then in a few Hours the Clouds will grow black, and be condensed into Rain, sudden and sharp, if in the Summer, but settled and moderate if in Winter.

THE old *English* Rule published in our first Almanacks agrees exactly with our Author's Observation.

If red the Sun begins his Race,
Be sure that Rain will fall apace.

IF the Reader would see this elegantly described, the Master of Poets hath it thus.[\[a\]](#)

Above the Rest, the Sun, who never lies,
Foretels the Change of Weather in the Skies;
For if he rise unwilling to his Race,
Clouds on his Brow, and Spots upon his Face,
Or if thro' Mists he shoots his sullen Beams,
Frugal of light, in loose and straggling Streams,
Suspect a drizzling Day and southern Rain,
Fatal to Fruits and Flocks, and promis'd Grain.

II.

If cloudy, and it soon decrease. } Certain fair Weather.

I Conceive the Reason of this to be, that the Vapours being then specifically lighter than the Air, are still rising upwards, in which they are assisted by the Heat of the Sun Beams, agreeable to the Notion of Dr. *Derham*, who observes, that after much cloudy Weather, it is always fair before it rains, because the watery Vapours are not condensed till they reach the cold upper Region, agreeable to the common *English* saying,

The Evening red, and Morning grey,
Is a Sign of a fair Day.

IT is also an Observation, of *Pliny's* in his natural History.[\[b\]](#)

SI ab ortu solis repellentur Nubes, & ad occasum abibunt, *Serenitatem* denunciabunt,

That is,

IF at Sun rising the Clouds are driven away, and retire as it were to the West, this denotes fair Weather.

THERE is an old Adage to this Purpose, which, because it is very prettily expressed, deserves our notice, viz.

A red Evening and a grey Morning,
Sets the Pilgrim a Walking.

In *French* thus.

Le rogue Soir, & blanc Matin;
Font rejouvir le Pelerin.

The *Italians* say the same.

Sera rosa, & nigro Matino;
Allegra il Peregrino.

III.

CLOUDS *Small and round, like a
Dappley-grey, with a North-Wind.* } **Fair Weather for
2 or 3 Days.**

THIS is differently expressed by other Authors. My Lord *Bacon* tells us, that if Clouds appear white, and drive to the *N. W.* it is a Sign of several Days fair Weather.

OUR old *English* Almanacks have a Maxim to this Purpose.

If woolly Fleeces spread the Heavenly Way,
Be sure no Rain disturbs the Summer Day.

AND *Pliny* to the same Purpose.[\[c\]](#)

SI Sol oriens cingetur Orbe, & postea totus defluxerit æqualiter, *Serenitatem* dabit.

That is,

IF the rising Sun be incompassed with an Iris, or Circle of white Clouds, and they equally fly away, this is a Sign of fair Weather.

THERE is another *English* Proverb worth remembering.

In the Decay of the Moon,
A cloudy Morning bodes a fair Afternoon.

IV.

Large like Rocks.——Great Showers.

I N the old Almanacks we have this Sign of the Weather thus expressed.

When Clouds appear like Rocks and Towers,
The Earth's refresh'd by frequent Showers.

THE Reason of this seems to be, that the watery Vapours are then condensed, or condensing, which gives them this rough and ragged Appearance, and as soon as the thin Films that retain the Water are broke by this Pressure, these heavy Clouds descend in Rain.

THESE Observations, as well as some that follow, are agreeable to all Climates, which is the Reason that they appear in so many different Authors, and have been taken notice of in so many Ages. This however does not at all diminish the Credit, or the Merit of our Shepherd's Observations, who certainly drew them not from Books, but from his own Experience, and therefore their agreeing so well with the Rules of other great Masters, ought to establish his Authority in such Cases as are not supported by alike concurrence from ancient or modern Writers, the Testimony of Nature is always sufficient Evidence.

V.

If small Clouds increase—Much Rain.

THIS and the following Observation cannot well be understood, without giving some Account of Clouds in general. The Atmosphere is supposed to extend itself about five Miles round this Globe of Earth, and within that Space move all kind of Vapours exhaled by the Sun's Force, or protruded by the subterraneous Heat. The ascending of these Vapours into the Air, depends upon many things, and therefore as different as its Causes; for instance, their ascent depends in the first place on the degree of Heat with which they are drawn up or forced out; next upon the Lightness of the Vapours themselves; thirdly, on the Density or Rarity of the Air through which they pass; and lastly, on the Force and Direction of the Winds, which they encounter in their Passage.

ACCORDING to the Nature of these Vapours, and the Circumstance attending their Passage, they appear to us differently below. For if they be extremely subtile they mount very high, and there, according to the Sentiment of Sir *Isaac Newton*, form by Refraction the Azure, or blue Colour, that over-spreads the Sky in serene Weather. Clouds, while they remain visible, do not rise above the Height of a Mile; and we always observe, that the highest are of a very light Colour, and hardly seen. If, therefore, small Clouds increase, it shews, that the Disposition of the Air is such, as that these Clouds cannot rise therein, either from their own Weight, the want of a protrusive Force, or from the falling of the Wind, which in cloudy Weather is always a Sign of Rain.

VI.

If large Clouds decrease——Fair Weather.

THE same kind of Reasoning accounts very clearly for this Prognostick, since it shews, that the Vapours are either exhaled by the Sun's Heat, or are driven off by Winds, and so resolved into smaller Clouds, capable of ascending higher in the Atmosphere; all which are Circumstances that secure us from Rain, and afford us a certainty of fair Weather.

IT is, however, to be observed, that large black Clouds are frequently, in a Summer Evening, melted into Dews; and this much more frequently happens in the Autumn, because the Evenings are then cooler, and the Vapours more easily condensed for that Reason. In all Observations of this Sort, there is a great degree of Prudence and good Sense required to apply them, and hence it very frequently happens that such Observations are condemned as treacherous and abusive, merely because those who would employ them want the Sagacity which is requisite to understand them clearly.

VII.

MISTS. *If they rise in low Ground and soon vanish.* } Fair Weather.

THIS is a sure Sign and very well expressed, that is, clearly, and, in few Words, which is the Excellency of such Aphorisms. In order to be convinced of its good Sense and Certainty, we must consider a little what *Mists* are, whence they rise, and what becomes of them.

MISTS are, strictly speaking, uncompacted Exhalations, which while they fleet near the Earth are styled *Mists*, but when they ascend into the Air, are called *Clouds*. If therefore, rising out of low Ground, they are driven along the Plain, and are soon lost to the Sight, it must arise from some of these Causes. That there is an Air abroad sufficient to divide and resolve them, or the Heat of the Sun has been strong enough to exhale them, that is, to rarify them, so as to render them lighter than the Air through which they were to pass. Whichever way this happens the Maxim remains unimpeached.

VIII.

If they rise to the Hill-tops. } Rain in a Day or two.

WHEN Mists are very, heavy in themselves, and rise only by the Action of that protrusive Force, exerted by the subterranean Fire, they can rise no higher than where the Gravitation becomes superior to that protrusive force, for then they descend again by their own Weight, and this occasions the Appearance mentioned in the Observation of their hanging upon Hill-tops, where they are very soon condensed, and fall down in Rain.

THERE was formerly a very idle and ill grounded Distinction between moist and dry Exhalations, whereas in Truth all Exhalations are moist, or in other Words are watery Steams thrown off by Bodies respectively dry, and the former Distinction was invented only to solve these Phænomena of which we have been speaking, that is, the Mist rising and, dispersing without Rain, and the Mist condensed and resolved into Rain, which as I have shewn may be much better explained without any such Distinction.

IX.

A general Mist before the Sun rises, near the full Moon. } Fair Weather.

THIS is a general and a very extensive Observation, which enables us to judge of the Weather for about a Fortnight, and there is very great Reason to believe that it will very rarely deceive us. In order to convince the Reader of this, it will be necessary to explain, as far as we are able, the Causes of this.

MISTS are observed to happen when the Mercury in a Barometer is either very low or very high. They happen when it is high after the Region of the Air has continued calm a good while, and in the mean time a great Abundance of Vapours and Exhalations have been accumulated, making the Air dark by their quantity, and the disorderly Disposition of their parts. They happen when the Mercury is low, sometimes because the Rarity of the Air renders it unable to sustain the Vapours, which therefore descend and fall through it.

BUT none of these Cases agree with the Observation at the Head of the Page, and therefore to form a true judgment of the Weather, we must distinguish between them and the Case which explains the Observation.

X.

If in the new Moon.—Rain in the old.

WHEN Exhalations rise copiously from the Earth into the Region of the Air, and the Air itself is in a proper Disposition, they ascend to a great Height, and continue a long Time before they are condensed, which accounts very clearly and philosophically for the Interval of fair Weather between the rising of these Mists, and their falling down again in Showers. Their ascending about Sun-rise is a Proof that the Air is thin, but at the same time of a Force sufficient to sustain them, since if the Mists were not specifically lighter than the Air itself they could not ascend.

WHEN the Moon is at the full, and such Exhalations rise plentifully, the time necessary for them to float in the Atmosphere, before they are condensed into Clouds and Rain, extends, generally speaking, beyond the Period of that Moon, and therefore the Observation directs us to expect *fair Weather*.

XI.

If in the old.——*Rain in the New.*

BUT as it is observed not only in this Climate, but all the World over, that great Changes of Weather happen near the Change of the Moon, it follows that this is the Season when these Exhalations that ascend so plentifully at Sun-rising are condensed, and consequently is the Season when we ought to expect *Rain*.

IF therefore the Exhalations rise in the new Moon, it is a Sign that the Air is in a fit disposition to sustain and support them for a considerable Time, and therefore we have Reason to expect that they should continue floating till the next regular change of Weather, that is, till the old of the Moon, or rather till towards the next Change, and therefore the Observation is very cautiously and very properly worded, directing us to expect Rain *in* the old, and *in* the new, and not *at* the old or new, because it is observed that these Changes of Weather happen not exactly *at* the Change of the Moon, but a Day or two before or after, of which the Reader will meet with many Examples in Captain *Dampier's* History of Winds and Storms at Sea.

XII.

WINDS. *Observe, that in eight Years' Time there is as much South West Wind, as North East, and consequently as many wet Years as dry.*

THIS must be allowed a very extraordinary Aphorism from a Country Shepherd, but at the same Time it is very agreeable to the Observations of Dr. *Hooke*, Dr. *Derham*, Dr. *Grew*, and other able Naturalists, who with unwearied Pains and Diligence have calculated the Quantity of Rain falling in one Year, and compared it with what fell in another. Lord *Bacon*, that Honour to our Nation and the Age which produced him, informs us, that it was an old Opinion there was a total Revolution of the Weather once in forty Years, and wishes it was inquired into. I cannot tell whether this has ever been done or not, but I think there is good Reason to conclude that there is a natural Balance established of wet and dry Weather, as of Light and Darkness, Heat and Cold, and other such like Variations.

IT may not be amiss to caution the Reader against a Mistake into which the Manner of this Rule being stated may easily lead him. It is this, that South West Winds cause Rain, and North East Winds fair Weather, which however is not a Thing clear or certain by any means. This indeed is true, that South West Winds and Rain, North East Winds and fair Weather come together, generally speaking. But the Question is, which causes the other, and a more difficult Question cannot easily be stated, because there seems to be Facts on both sides. South West Winds seldom continue long without Rain, this seems to prove the affirmative, but on the other Hand, when in hard Weather, Rain begins to fall, the Wind commonly veers to the South West, this looks as if the Rain caused the Wind.

BUT to keep close to the Shepherd's Observation. There is one Thing seems strongly to confirm it, which is this, that in any given Place the Quantity of Rain one Year with another is found to be the same by Experience, according to which the following Table has been calculated, for the mean Quantity of Rain falling one Year with another in those Places that are mentioned, and on this Proportion the other seems to be founded.

At Harlem	24 Inches
Delf	27
Dort	40
Middleburg	33
Paris	20
Lyons	37
Rome	20
Padua	37½
Pisa	34¼
Ulm	27
Berlin	19½
In Lancashire	40
Essex	19½

XIII.

When the Wind turns to North East, and it continues two Days without Rain, and does not turn South the third Day, nor Rain the third Day, it is likely to continue North East for eight or nine Days, all fair; and then to come to the South again.

I N my Opinion this and the subsequent Remarks depend entirely upon Observations, and may serve rather to found an Hypothesis, than seem be deduced from one. That the Variations of the Wind depend on certain Causes, and may consequently be reduced to Rules, is highly probable, and such Observations as these render it in a manner certain. But to explore these Causes, and to explain them in such a manner as to account for these Phænomena in a satisfactory manner, requires not only great Sagacity but much Experience, and many Years' Observation, which, however, considering the great Benefits that would result to Mankind from establishing such a THEORY, would be Time well bestowed.

WE may however easily conceive that a constant North East Wind must be accompanied with fair Weather. For whatever the causes of Winds may be, yet on this side the Equator, a strong and settled North East always buoys up the Clouds and keeps them suspended. This has been long observed by, and passes for a settled point amongst Seamen. The Reason of it however cannot be so easily assigned, at least a satisfactory Reason, for as to Suppositions, every fanciful Man can furnish them at Pleasure.

THIS, as well as the following Observations, very plainly and clearly prove, that in this Part of the World fair Weather attends one Wind, and wet another, but which is the Cause and which the Effect, or whether both are not the Effects of some other Cause, I pretend not absolutely to determine. But inasmuch as it is certainly known, that Rains attend in other Climates those Winds that are here attended with fair Weather, it seems more agreeable to suppose that rainy Weather is occasioned chiefly by West Winds, because loaded with moist Vapours from the Sea.

XIV.

If it turn out again out of the South to the North East with Rain, and continues in the North East two Days without Rain, and neither turns South nor rains the third Day, it is like to continue North East for two or three months.

The Wind will finish these Turns in three Weeks.

THIS Observation is of the same nature with the former, and is plainly deduced from long experience. Our Author seems to contradict himself in saying that these Winds finish their Turns in three Weeks, but his true Meaning certainly is, that they are \about three Weeks in turning from the South to the North East again. Some very great men have laid it down as a thing certain, that the Variations of the Wind are to be accounted for by the Alteration of the Balance of the Air, occasioned by the different Effects of Heat and Cold; but other Writers again insist very copiously on the Effects which Winds have upon the Air, and thus confound us in a Circle of Causes and Effects, whence it is plain that they do not thoroughly understand the Subject themselves, and therefore it is no Wonder that they are not able to explain it to others.

IN some Parts of the World, and especially between the Tropicks, the Winds are regular, and therefore our Philosophers seem to talk more rationally about them. But in our Northern Countries the Alterations of the Wind are so frequent, sudden, and often so little agreeable to the Season, that such general Reasonings will by no Means serve to explain them. It is however

very reasonable to suppose that the same general Cause prevails here as between the Tropics, but with less Certainty, because the Power of the Sun is not so great, and the Determinations of the Winds depend on the Situation of Mountains, Rocks, and Woods, which direct the Air driving against them into certain Courses, so that it is impossible to explain, or indeed to judge of the Course of the Winds till the Country is thoroughly known, and all those Eminences that can affect the Winds are well considered.

FROM these Reflections the Value of our Shepherd's Observations will clearly appear. He was not Philosopher enough to talk in this Style, but by a long and steady Attention he came to know, experimentally, what perhaps few Philosophers, with all their Sagacity, would have been able to have found out.

XV.

S. W. WINDS. *After a northerly Wind for the most Part two months or more, and then coming South, there are usually three or four fair Days at first, and then, on the fourth or fifth Day, comes Rain, or else the Wind turns North again, and continues dry.*

THIS is likewise a very judicious and very useful Observation, and yet it is not a difficult matter to account for it. It is a common Observation, and a very true one, that there is usually fair Weather before a settled course of Rain. The Winds that bring the dark rainy Clouds that obscure the Sky, and cause dull cloudy Weather, often raise these Vapours to such a height, that they are attracted into the cold Region above our Sight, till being condensed there, they fall down upon us again in Snow or Rain, according to our Author's Observation.

BUT if, after a seeming Tendency to Rain, there follow several Days of fine Weather, it is a certain Indication that the Temper of the Air is altered, and that these Vapours had been driven off before they had time to condense, which is confirmed by the Change of the Wind on such Occasions.

ALL these Observations are to be understood in a proper Latitude, and not strictly and according to the very Letter. For Rain may fall the sixth or seventh Day, or the Wind may change the second or third. Besides, a Man who would make use of these Observations in the Country, must consider attentively the Situation of the Place where he lives, the bearing of the Sea, Marshes, Ponds, Lakes, Woods, Mountains, Rocks, &c. For without making

proper Allowances for these, all such Observations on the Weather will be apt to fail him.

XVI.

If it return to the South within a Day or two without Rain, and turn Northward with Rain, and return to the South, in one or two Days, as before, two or three Times together after this Sort, then it is like to be in the South, or South West, two or three Months together, as it was in the North before.

The Winds will finish these Turns in a Fortnight.

THIS may appear a little perplexed to an ordinary Reader, but a little Attention will make it very clear and plain; and whoever considers what mighty Uses may be made of the Foresight of Weather for a Month or two, will not think this Labour ill bestowed. I must confess I look upon these three Rules in Relation to the Wind as the most useful in the whole Collection. Especially to Farmers and Country People, to whom they are of the greatest Consequence.

BUT it is a common Thing for such People to say, what Certainty is there that these Rules will prove true, what Probability is there that the Wind should continue so long in one Quarter, and then so long in another, how shall we be satisfied that there is any truth in this; or, if we cannot be satisfied as to the Truth of it, why should we depend upon any such like Observations?

TO this I answer, that they may have reasonable Satisfaction given them on this Head. Some of our great Naturalists, who had kept Journals of the Weather for many Years, have found that the same Wind blows every Year very near the same number of Days, and that there is a regular Continuance of different Winds annually in every Country. For Instance,

At *Utrecht* they blow thus,

The N. Wind	42 Days.
The N. W.	33
The W.	77
The S. W.	58
The South	33
The S. E.	26
The E.	53
The N. E.	<u>43</u>
	365

IT is a Thing plain to every Capacity, that a Journal or Diary of the Winds may be kept any where, and if from such a Journal it appears that a given Wind blows for a certain Number of Days, then it follows, that if these can be determined with Certainty, the Time of their blowing may also be determined, at least with great Probability, which is as satisfactory an Answer as can be justly expected, because it shews that there is just and rational Ground for confiding in such Observations, when confirmed by long Experience.

XVII.

Fair Weather for a Week, with a Southern Wind, is like to produce a great Drought, if there has been much Rain out of the South before. The Wind usually turns from North to South, with a quiet Wind without Rain, but returns to the North, with a strong Wind and Rain; the strongest Winds are when it turns from South, to North by West.

N. B. *When the North Wind first clears the Air (which is usually once a Week) be sure of a fair Day or two.*

OBSERVATIONS of this Nature upon Winds have employed the ablest Heads in all Ages. *Pliny* the great Naturalist has left us a great deal upon this Subject, which plainly proves that it has been the Opinion of the ablest and wisest Men that Study and Experience might reduce even Things of such seeming Incertainty under stated Rules, and within the Bounds of a regular System. For Instance he tells us.

"IN *Africa* the South Wind is serene, the North East cloudy. All the Winds have their Turns. To judge rationally of their Changes, the fourth Day of the Moon is to be regarded.—The South Wind blows stronger than the North East, because the former rises from the Bottom, whereas the latter comes from the Surface of

the Sea. It is for this Reason that those Earthquakes are most dangerous that follow after a South Wind."

IN order to understand this Notion of *Pliny*, we need only advert to the Account given us by the Reverend Mr. *Robinson*, in his natural History of *Westmoreland*, which is exceedingly curious, and well worthy of the Reader's perusal. This ingenious Gentleman is of Opinion that Winds have their original from the Sea, of which he gives the following very probable Account.

"IT, that is, the Wind, proceeds from vast swarms of nitrous Particles arising from the Bottom of the Sea, which being put into Motion, either by the central Fire, or by the Heat and Fermentation which abound in this great Body of the Earth; and therefore the first Commotion excited by the said Fermentation, we call a *Bottom Wind*, which is presently discovered by Porpusses and other Sea Fish, which delight in sporting and playing upon the Waves of the Sea, and by their playing give the Mariners the first Notice of an approaching Storm.

"WHEN these nitrous swarms are risen towards the Surface of the Sea in a dark Night, they cause such a shining light upon the Waves, as if the Sea was on fire. And being delivered from the brackish Water, and received into the open Air, those fiery and shining Meteors which fix upon the Masts and Sides of the Ships, and are only nitrous particles condensed by the circumambient Cold, and like that which the Chymists call Phosphorus, or artificial Glow-worm, shine and cast a Light but have no Heat: This gives the Mariners the second Notice that the Storm is rising, for upon the first breaking out of the Wind, the Sea begins to be rough, and the Waves swell and rise, when at the same time the Air is calm and clear.

"THIS boiling Fermentation of the Sea causes the Vapours to rise, which by the Intenseness of the circumambient Cold are condensed into thick Clouds, and fall down in Storms of Wind and Rain, first upon the Sea, from whence they rose, and then the attractive Power of the Mountain-cold, by a secret Magnetism between Vapour and Cold, attracts the waterish Vapours, intermixt with nitrous Particles, to the high Tops of Mountains and Hills, where they hang hovering in thick Fogs and waterish Mists, until the atmospherical Heat rarefies the nitrous Part of the Fog (which is always uppermost, and appears white and translucent) into brisk Gales of Wind, and the Intenseness of atmospherical Cold having attracted the Vapours into the colder Regions of the Air, where being condensed into Clouds, the Wind breaks, dissipates, and drives them before it, till they fall down in Rain, and water the Surface of the Earth. And this seems to be the Reason why, in *Egypt*, and those level Countries where they have no Mountains, they have little Wind and less Rain."

XVIII.

CLOUDS. *In Summer or Harvest, when the Wind has been South two or three Days, and it grows very hot, and you see Clouds rise with great white Tops like Towers, as if one were upon the Top of another, and joined together with black on the nether Side, there will be Thunder and Rain suddenly.*

WE may very easily account for this Observation, because in Fact the Signs here mentioned are no other than Nature's apparatus for a Storm of Thunder and Lightning, which will be perfectly understood by attending a little to the Causes of these Meteors. Lightning is a great flame, very bright, extending every way to a great distance, suddenly darting upwards, there ending, so that it is only momentaneous. The Matter which produces the Fire, is the Oil of Plants, attenuated by the heat of the Day, and raised on high. Then whatever has exhaled from the Earth that is sulphureous or Oily, which is dispersed up and down in the Atmosphere, and is not continuous, is set on Fire by Turns, and the Flame dilates itself as far as the Tract of that Exhalation reaches. Some other Substance pendant and floating in the Air meets with this also, with which it excites an effervescence, takes Fire and flashes along with it. Thunder is another bright Flame, rising on a sudden, moving with great Velocity through the Air, according to any Determination upwards from the Earth horizontally, obliquely, downwards in a right Line, or in several right Lines as it were in serpentine Tracts joined at various Angles, and commonly ending with a loud Noise or Rattling.

IT is observed that it thunders most when the Wind blows from the South, and least when it blows from the East. The great Principle of Thunder is Sulphur, as is evident from the Smell it leaves behind it; but in order to occasion such an Explosion, there must be other Ingredients mixed therewith, especially Nitre, of which the Air is always full, besides other Things, of which it is impossible to give any Account. The Tracts of this Sort of Matter fly about in the Air, and are as it were Lines of Gunpowder, and as in the firing of that Powder, the Fire begins at one End, and pursuing its Aliment proceeds to the other Extremity, and so the whole Mass of Powder is fired; we may from thence account for the Phænomenon of Thunder. For in like Manner those inflamed Tracts which are suspended in the Air, flash from a Flame that runs from one Extreme to the other, wherever the Vein of Nourishment leads it. Hence those Rays of Thunder, which seem to be brandished through the Air, and sometimes to be split in two or more Tracts, and sometimes to return back, at other Times to be projected in Lines that are joined by various Angles, and this only because the Flame meets with Tracts lying in various Situations that cohere one with another. Therefore Thunder seems now to run horizontally, now from above downwards, now upwards from the Earth, for if the Matter of Thunder pressing out of the Earth is enflamed near the Ground, the Flame darting upwards, the Thunder will seem to be projected out of the Earth. If the same Tract be set on Fire at its upper end, the Flame will move downwards, and the Thunder will seem to descend out of the Sky.

HENCE we easily understand how it comes to thunder oftener in one Place than another, but most frequently in those where the Soil produces odoriferous Herbs, and abounds with Sulphur, and where the People are much exposed to the extreme Heat of the Sun. Thunder is less frequent in Places where there are few odoriferous Herbs, very little Sulphur, or where the Climate is watery and moist. For Instance, it thunders very much in *Italy* and *Sicily*, and very rarely in *Egypt*, and the adjacent Countries. If it be demanded how it comes to thunder in the midst of the Ocean? The Answer is easy, because from the Bottom of the Ocean vast Tracts of sulphureous

Matter are cast up through the Waters; as it happens to spring Waters in several Places, the Streams of which will take Fire from a lighted Candle. For sulphureous Exhalations bursting out together with the Waters, the fulmineous Matter in the Air is set on Fire when it meets with Exhalations or Vapours with which it can excite a vehement Effervescence. It is very clear from this Account, that the Clouds mentioned at the Top of the twenty-eighth Page are thunder Clouds, or Clouds big with the Materials of Thunder.

XIX.

If two such Clouds arise, one on either Hand, it is Time to make haste to shelter.

AS this Observation is of the same Nature with the former, we shall continue our Remarks. The Reason why it seldom thunders in Winter is, because the exterior Parts of the Earth are so contracted by the cold Snow and Ice, that Sulphur cannot perspire in any great Quantities, but as soon as the Earth begins to be opened by the Sun in the Spring, something expires in the Month of *April* which takes Fire. But by the greater Heat of the Sun penetrating deeper into the Earth, the Cortex is more opened in *May*, and now there is a more copious Expiration of the fulminating Matter, and whatever was collected and shut up in Winter, is now released and snatched up in the Air, and thence proceeds the most frequent Thunders in the Month of *May*, and chiefly when a very hot Day or two has gone before. A less Quantity of the same Matter remains in the upper Cortex of the Earth for the Month of *June*, but in the mean Time a Stock arises out of the deeper Bowels of the Earth, which is attenuated and prepared, so that by the very fervent Heat of *July* it is elevated, as it were in heaps, and set on Fire. Hence Thunder is as frequent in *July* as in *May*. And the Heat decreasing in the succeeding Months, the Exhalation of the fulminating Matter out of the Earth is more sparing, and thence, also, the thunder is less frequent, till in October, and the other winter months, the earth is bound up with us, and hardly expires any more. Hence we see why it very seldom thunders when the northerly winds blow; for these winds constringe the earth with their cold, and so hinder the fulminating matter from bursting forth; and when they are burst forth and floating in the air, they hinder their effervency. But

on the contrary, when the warm and moist south winds blow, which open every thing, the earth likewise is opened, and abundance of fulminating matter is expired and ascends on high, which is there easily inflamed.

AS the flame runs very swiftly, it seems to carry along with it particles, which it could not so easily set on fire, and when any of these particles are drawn together, and heated to a certain degree, they at last take fire, with a sudden and great explosion, and thereby produce what we call a thunder Clap. Now, though this be only a single sound, yet it is often heard in the form of a great murmuring noise of a long continuance; sometimes for thirty or forty seconds, because of its various repercussions by the clouds and terrestrial obstacles. Hence it is, that in vales, which are surrounded by mountains of a different Height, there is a terrible and long continued Bellowing of thunder Claps. Whereas for one Explosion it has been observed that there is but one Clap. Yet however if the Flame set Fire to two, three, or more fulmineous Tracts, each of them at last will end in a Clap, and thus several Sounds may be heard together, or quickly succeeding one another.

XX.

*If you see a Cloud rise against the Wind or side Wind,
when that Cloud comes up to you, the Wind
will blow the same Way that the Cloud came.
And the same Rule holds of a clear Place,
when all the Sky is equally thick, except one
clear Edge.*

THIS seems to arise from hence, that Wind being nothing more than Air in motion, the Effects of it first discover themselves above, and actually drive such Clouds before them. This was long ago observed by *Pliny*. When Clouds, says he, float about in a serene Sky, from whatever Quarter they come, you may expect Winds. If they are collected together in one Place, they will be dispersed by the approach of the Sun. If these Clouds come from the North East, they denote Winds; if from the South great Rains. But let them come from what Quarter they will, if you see them driving thus about Sunset, they are sure signs of an approaching Tempest.

IF the Clouds look dusky, or of a tarnish silver Colour, and move very slowly, it is a Sign of Hail. But to speak more plainly, those very Clouds are laden with Hail, which if there be a Mixture of Blue in the Clouds will be small, but if very yellow, large. Small scattering Clouds that fly very high, especially, from the South West, denote Whirlwinds. The shooting of fallen Stars through them, is a Sign of Thunder. We meet with many Observations of this sort in our old Writers on Husbandry, and we have abundance of Proverbs relating to this Subject which are worth observing, and the rather, because most of them are not peculiar to our Language only, but common to us with many of our Neighbours. It is the Remark of Lord *Bacon*, and a

very judicious Remark too, that Proverbs are the Philosophy of the common People, that is to say, they are trite Remarks founded in Truth, and fitted for Memory. I must confess that there are some of them that seem either false, or of no great Consequence, but then I am apt to suspect, that by various Accidents we have lost their true Meaning, or else, that in length of Time, they have been altered and corrupted, till they have little or no meaning at all.

I cannot help taking Notice in Regard to the Rule before us, that Captain *Dampier* tells us in the *East-Indies*, they have always Notice of a Tuffoon by the Skies being first clear and calm, and then a small white Cloud hanging precisely in the Point from whence the Storm comes, where he observes that it remains sometimes twelve Hours or more, and adds, that as soon as it begins to move, the Wind presently follows it. When Sir *John Bury*, who died an *English* Admiral, had the Command of a small Frigate in the *West-Indies*, he escaped a Hurricane in the *Leward* Islands by taking the Advice of a poor Negro, who shewed him a small white Cloud at a Distance, and assured him that when it came to the *Zenith*, the Hurricane would infallibly begin, as indeed it did.

XXI.

Sudden Rains never last long: But when the Air grows thick by Degrees, and the Sun, Moon, and Stars shine dimmer and dimmer, then it is like to rain six Hours usually.

RAIN is, properly speaking, a Multitude of small watery Drops, falling from the upper Air at different Seasons. When the upper Regions become cold of a sudden, the watery Clouds are condensed and fall in hasty Showers. It is observed that mountainous Countries have most Rain, and the Reason seems to be the Winds driving the Clouds against the Rocks and Hills, and thereby compressing them in such a Manner, that they are immediately dissolved, and fall as it were at once. This is the Reason that in *Lancashire* there falls twice as much Rain as in *Essex*, and it is probably from the same Cause, that in the Ocean, over-against the mountainous Coast of *Guinea*, showers sometimes fall as it were by Pailfuls.

THIS Observation of our Shepherd is very just and reasonable, and I dare say will hardly ever fail such as observe it. The Dimness of the Stars and other heavenly Bodies, is one of the surest Signs of very rainy Weather. It is likewise to be observed that when the Stars look bigger than usual, and are pale and dull and without Rays, this undoubtedly indicates that the Clouds are condensing into Rain, which will very soon fall; and it has been observed that when the Air grows thick by Degrees, and the Light of the Sun lessens so as not to be discerned at all, and again when the Moon or Stars have the same Appearances, a continued Rain for at least six Hours is sure to follow.

TO be the better informed in such Cases, it is best to have Recourse to a variety of Signs, for it is not only the Clouds and Sky, or the Sun, Moon, and Stars, that gives us previous Notice of rainy Weather, but almost every Thing in the Creation, and Vegetables particularly. As for Instance, the Pimpernel, which is a very common flower, shuts itself up extremely close against rainy Weather. In like manner the Trefoil swells in the Stalk against Rain, so that it stands up very stiff, but the Leaves droop and hang down. Even the most solid Bodies are affected by this Change of the Atmosphere, for Stones seem to sweat, and Wood swells, the Air driving the moist Particles with which it is filled into the Pores of dry Wood especially, makes it swell prodigiously, and this is the Reason the Doors and Windows are hard to shut in rainy Weather.

THIS is so true, that there has been a Method found of dividing Mill-stones by the mere Force of the Air, which is done in this Manner. They divide a Block of this kind of Stone as big as a large rolling Stone, into as many Parts as they design to make Millstones, and in the Circles where this Block is to be divided, they pierce several Holes, which they fill with allow Wood dried in an Oven, and expose the Stone to the Air, in moist Weather; when the Wood swells to such a Degree as to split the Stone as effectually, as if it was by iron Wedges driven by Sledge-Hammers. This curious and extraordinary Method of dividing Mill-stones is related by the famous Mr. *Ozanam* of his own Knowledge.

XXII.

If it begin to Rain from the South, with a high Wind for two or three Hours, and the Wind falls, but the Rain continues, it is like to rain twelve Hours or more, and does usually rain till a strong North Wind clears the Air. These long Rains seldom hold above twelve Hours, or happen above once a Year.

THIS depends entirely upon Observation, and Experience shews us that whenever the Wind falls, Rain follows. It has been likewise observed, that when the Wind changes often there fall heavy Rains. All these Alterations in the Atmosphere, are less observed by Men than by Animals, for two Reasons. The first is, that we live much within Doors, by which they are less obvious to us, and it is for this Reason that the Husbandman, Seamen, Fishermen, but above all Shepherds, who are more in the open Air than other Men, are better acquainted with, and more able to distinguish and judge of the Signs of the Alteration of the Weather, than those who live altogether within Doors, or go out but seldom. Another Reason is our having so many Things to mind, which takes off our Thoughts, and renders us less attentive to the Signals which would give up Notice of such Alterations. It is for this Reason that we ought to serve ourselves of that Sort of Instinct which Nature has given to other Animals, and which as it is a Gift of Nature, is in a Manner infallible.

THUS if small Birds prune themselves and duck and make a shew of washing. If Crows make a great Noise in the Evening, if Geese gaggle more than usual, these are all Signs of Rain, because these Animals love wet

Weather, and rejoice at the approach of it. On the other Hand, if Oxen lie on their Right Sides, look towards the South, and lick their Hoofs, if Cows look up in the Air, and snuff it, if Asses bray violently, and if Cocks crow at unusual Hours, but especially when a Hen and Chickens crowd into the House, these are sure Signs of Rain.

INSECTS also are very sensible of such Changes of Weather. Frogs croak more than ordinary, Worms creep out of the Ground, Moles throw up more Earth than usual, because such Weather is more agreeable to them; Hornets, Wasps, and Gnats, sting more frequently against wet Weather than in fair. Spiders are restless and uneasy, and frequently drop from the Wall, the humid Air getting into their Webs and making them heavy. But the surest and most certain Sign is taken from Bees, which are more incommoded by Rain than almost any other Creatures, and therefore, as soon as the Air begins to grow heavy, and the Vapours to condense, they will not fly from their Hives, but either remain in them all Day, or else fly but to a small Distance.

XXIII.

If it begins to rain an Hour or two before Sun-rising, it is like to be fair before Noon, and so continue that Day, but if the Rain begin an Hour or two after Sun-rising, it is like to Rain all that Day, except the Rainbow be seen before it rains.

THIS is a short, clear, and easy Observation, and therefore I shall not dwell long upon it, but rather entertain the Reader with a few Observations on the Rainbow. Whenever it appears, things are thus circumstanced. The Spectator has the Sun behind him, and Clouds with the Bow in them before him. Sometimes there are two and even three Bows seen, but this is very rare. The Colours in the Bow are ranged in this Order, viz. Violet, Purple, Blue, Green, Yellow, Orange, Red. After a long Drought the Bow is a certain Sign of Rain, if after much Wet fair Weather. If the Green be large and bright it is a Sign of Rain, but if the Red be the strongest Colour, then it denotes Wind and Rain together. If the Bow breaks up all at once there will follow serene and settled Weather. If the Bow be seen in the Morning small Rain will follow. If at Noon, settled and heavy Rains; if at Night, fair Weather. The Appearance of two or three Rainbows shews fair Weather for the present, but settled and heavy Rains in two or three Days' Time.

LUNAR Rainbows are sometimes, but very seldom seen, they are extremely beautiful, but much less than those that appear in the Day time, and a yellow, or rather a straw Colour prevails most. As they happen so seldom, they cannot well be reckoned amongst the Signs of Weather. But now, after speaking of so many different Methods of judging when rainy

Weather will be of a short or long Continuance: Give me leave to describe two or three Instruments easily made, which will shew the Alterations of the Weather certainly, constantly, and early enough for most Uses.

THERE were some Years ago a Sort of Toys sold, with a Man and a Woman so fixed before the Door of a House, that at the Approach of wet Weather the Woman entered it, and when the Weather grew fair the Man. This was done by the Help of a Bit of Catgut, which shrinks in wet Weather, and stretches again when it is fair. This appears better by a Line and Plummet, especially if the Line be made of good Whipcord, that is well dried, for then if it be hung against a Wainscot, and a Line drawn under it exactly where the Plummet reaches, in very moderate Weather it will be found to rise above it before Rain, and to sink below when the Weather is like to become fair; but the best Instrument of all is a good Pair of Scales, in one of which let there be a brass Weight of a Pound, and in the other a Pound of Salt, or of Salt-Petre well dried, a Stand being placed under the Scale, so as to hinder its falling too low. When it is inclined to rain the Salt will swell, and sink the Scale, when the Weather is growing fair, the brass Weight will regain its Ascendancy.

XXIV.

SPRING and SUMMER. *If the last eighteen Days of February and ten Days of March be for the most part rainy, then the Spring and Summer Quarters are like to be so too: and I never knew a great Drought but it entered in that Season.*

IT is easy to discover by Observation whether this Rule be well or ill founded, that is to say, whether our Shepherd's Observation will serve for other Places or not, and where it will serve and where not. But it may not be amiss to remark that it is highly probable, or rather absolutely certain, that the Weather in one Season of the Year determines the Weather in another. For instance, if there be a rainy Winter then the Autumn will be dry, if a dry Spring, then a rainy Winter. Our Forefathers had abundance of odd Sayings upon this Subject, and some Proverbs for every Month in the Year, but I doubt they were indifferently founded, however there can be no Harm in observing them, in order to discover whether there be any thing in them or not.

Janiveer freeze the Pot by the Fire

If the Grass grow in *Janiveer*

It grows the worse for't all the Year.

The *Welchman* had rather see his Dam on the Bier

Than to see a fair *Februeer*.

March Wind and *May* Sun

Make Clothes white and Maids Dun.

When *April* blows his Horn
It's good both for Hay and Corn.
An *April* Flood
Carries away the Frog and her Brood.
A cold *May* and a windy
Makes a full Barn and a Findy.
A *May* Flood never did good.
A Swarm of Bees in *May*
Is worth a Load of Hay.
But a Swarm in *July*
Is not worth a Fly, &c.

XXV.

WINTER. *If the latter End of October and Beginning of November be for the most Part warm and rainy, then January and February are like to be frosty and cold, except after a very dry Summer.*

IT is very evident, supposing this Observation to be true, as I am pretty confident it is, that the Reason of it is to be sought in that Balance of the Weather which Providence has established. There is not only a Time to sow, and a Time to reap, but there is a Time also for dry and a Time for wet Weather, and if these do not happen at proper Seasons, they will certainly happen at other Seasons; for not only the Wisdom of Philosophers hath discerned, but their Experiments and Observations have put it out of doubt, that there is a certain Rule or Proportion observed between wet Weather and dry in every Country, so that it is nearly the same in every annual Revolution, neither is wet and dry Weather only, but hot and cold, open and frost, that are thus regulated, from whence we see, that when the Scripture represents to us God's settling Things by Weight and Measure, it speaks not only elegantly, but exactly. For we do not mean by Providence any extraordinary or supernatural Interposition of almighty Power, but the constant and settled Order established by the Will of that almighty Being which we commonly call Nature.

THERE is nothing easier than for vulgar Understandings to mistake the Meaning of Words, and by a Superstition natural to weak Minds convert, what they imperfectly understand into Notions that perplex and confound them. Hence it proceeds that in common Conversation one hears People

speak of Nature as of a Being, or a Kind of subordinate Deity, whereas in Reality the true Meaning of Nature is, that Order or Law which God has established in the Universe, and the Knowledge of Nature is no more than the Light we acquire by Study into the Connexion of those Laws. In this Sense Experience is a Kind of Revelation, that is to say, it is a Sort of Knowledge that comes to us from without, and is infallible in itself, we may indeed go on wrong and deceive ourselves in the Arguments we raise from it, but the Knowledge grounded upon Experiments never varies.

THIS is sufficient to shew us how much wiser a Thing it is to trust this Sort of experimental Knowledge, then to put any Faith in that Kind of idle Science which amused our Forefathers, and enabled Almanac Makers to delude and mislead them. It is true we use the Luminaries as well as they, but then we use them in a rational Manner, and do not pretend to impose this or that Sign upon other People, but barely set down our own Observations, which are to be examined and verified by the Experience of those to whom they are submitted. The Astrologer on the other Hand insists on what are not in Nature; the twelve Houses are a mere Invention, and so are all the Properties ascribed to the celestial Signs, and to the Planets; mere Dreams and Fictions devised by the Cunning to cheat and impose upon the Ignorant, and which had been long ago exploded if People had brought them to the only Test of which they are capable, I mean that of Experience; with which they never did, never will, and indeed never can agree: whereas the Rules given by our Shepherd, are such as we have shewn, suit perfectly well with Remarks of other studious Persons in all Ages.

XXVI.

*If October and November be Snow and Frost, then
January and February are like to be open and
mild.*

AS this Observation stands on the same Foundation with the last, we need not dwell upon it particularly, and therefore I shall proceed with my former Reflections. The only Way to be acquainted with Nature, is to study Nature. All Systems of human Invention that are not built upon Experiments, are sooner or later found to be false, because, to say the Truth, they are nothing better than ingenious Contrivances invented by the Wit of Man, to conceal his Ignorance. In order to account for what we behold, we must first of all take Pains to be well acquainted with the Fact, and not suffer ourselves to be led away by Opinion. In order to explain what I mean, I shall give an Instance. All the World knows that not only the vulgar, but the learned, were for many Ages in a constant Error about Corruption, and really believed that the Heat of the Sun, and even animal Heat produced Worms, Maggots, and other living Creatures. Many grave Writers carried the Thing farther, and told us of Rats, Mice, and other Creatures produced out of the Slime of the River *Nile*, by the Heat of the Sun in *Egypt*, which might very well pass for Truth among those who fancied they saw every Day something of the like Nature: I mean in the Corruption of Flesh and other Things, in which we behold Thousands of living Creatures.

AN *Italian* Philosopher destroyed this whole Doctrine at once, by a simple and easy Experiment. He exposed a Piece of raw Flesh in a glass Vessel well covered with Gauze to the Air and Sun, and found that it putrefied without producing any living Creatures. This shews how careful we ought

to be with Respect to Facts; for till this Experiment was made, no Body doubted that Vermin were bred by, as well as in putrefied Bodies; whereas we are now satisfied that the Heat of the Sun can no more produce a Worm or a Maggot, than a Horse or an Elephant. By the same Examination we might open the Way to Knowledge, by driving out a Multitude of other Errors. But the Humour of taking Things for granted without inquiring into them, and then endeavouring to account for them by dint of Reasoning, amuses us with a false shew of Wisdom, and encourages us to persist obstinately in the Maintenance of weak and foolish Notions.

TO apply this to the Subject of which we are treating. It is certainly a curious and a useful Thing to understand the Nature of the Weather, and to know how the Changes that happen in it come to pass. The Business is to find out the true Way of coming at this kind of Knowledge, and upon the Principles that I have advanced, it is very evident that the, only certain Way of coming at it is by Observation. This is a slow but a sure Method of arriving at Truth, and the Specimen here given us, of *one* Man's Observations, is enough to convince us that a little Diligence and Application would soon go a great Way towards forming a Body of such Observations as might enable us to understand the Weather thoroughly, and to predict its Changes and Alterations with a great Degree of Certainty. If we will not take this Pains, we must content ourselves with what hath been already discovered, or if our Conditions of Life exclude us from the Opportunities of making such Observations, it is certainly a right Thing to help ourselves by inquiring into, and reasoning upon such Observations as other People have made and to facilitate this as far as possible, I have taken the Pains to write this Commentary upon our Shepherd's Rules; which I hope will render them more useful, or at least secure them that Regard which they deserve.

THERE remains therefore nothing more for me to do in order to recommend these Observations, but to say somewhat with Respect to the Utility of the Alterations of the Weather in general, and in particular; in

order to satisfy the Reader that there is nothing of Chance or Accident in such Alterations, but that they are governed in every Respect by the same unerring Wisdom, that at first framed and constantly preserves the Universe. All Weathers are at sometimes reasonable, which shews that they are good in themselves, and only accidentally evil. We ought not to measure Things of a general Nature, by particular Rules. If by the Direction of Providence the Succession of Seasons be such, as that they turn to the good of Mankind in the whole; it is no Objection to, or Diminution of Providence that this Succession of Seasons should at different Times be injurious to certain Countries, because this may likewise be accounted for.

AS to Particulars we will begin with the Air, which is composed of Exhalations of all earthly Bodies, as well solid as fluid, as also of Fire, whether of the Sun or the Stars, or of earthly Bodies burnt, or of Fire breaking out from the Entrails of the Earth, and ascending, and though it be thus compounded, and hath swimming in it Multitudes of other Things, yet we find that it is perfectly wholesome, is the Spring of Motion, and of Life to Men, and all other Animals; so that though we cannot account for, and perhaps have not a Power of comprehending how such a mixed Body can be rendered salutary: yet since it is certain, that so it is, we have no Right to complain either of the evil Consequences that sometimes attend the Exhalations with which it is filled, or the Accidents that flow from the frequent Alterations that happen therein, because these have a visible Tendency to the general good, and are apparently necessary to the Preservation of the Universe, so that before we can have any Title to find fault, we must first shew that we are capable of understanding them in their full Extent, and as *this* is impossible, it follows *that* must be unreasonable.

BUT this appears still the more clearly, when it is considered, that all such Alterations may be shewn even from the Light of Reason to be generally useful, notwithstanding they sometimes appear troublesome and noxious. For Instance, such quick Streams of Air in Motion as we call Winds, though they sometimes swell into Storms and Tempests, yet are they of great

Benefit to Mankind, by purging the Air, and many other Conveniences. It is a Proverb at *Vienna*, *that if Austria be not windy it is sickly*; and this Saying is no less true in other Countries, for by consulting the History of the last great Plague that raged here in 1666, it will be found that there was in a Manner a dead Calm during the Time of the Sickness, and it is known in *Egypt*, where they have Plagues annually, that the Change of the Wind delivers them from that Evil. Add to this the great Use of Winds in Navigation, and reflect on the Benefits that accrue there from, and we shall see no Cause whatever to doubt that this Motion of the Air is a very wise Contrivance.

THE Condensation of Vapours, which is the Cause of Rain, is another great Benefit to the World, in as much as this is very probably supposed to be the Source of Fountains, Rivers, Lakes, and other Magazines of fresh Water, without which the Earth would be uninhabitable, and to which in a very great Measure its Fertility is owing. We ought likewise to remember that though this be in itself so clear, and at the same Time so certain, yet there are Countries in the World where it very seldom rains, as in *Egypt*, and others where it hardly ever rains, as in *Peru*; so that we see there is no raising general Doctrines upon this Subject, which ought to make us the more tender in disputing the Will of Providence, or repining when it happens to cross our own.

THE Uses of Snow are as great though less apparent, of which I shall mention but three. The first is its preserving Herbs and Grass in the Winter against the Severity of Frost; secondly, its supplying Water to Brooks and Rivers; and lastly, its furnishing the Earth with vast Quantities of Nitre, and thereby conducing greatly to its Fertility, and perhaps the same thing may be said of Frost, hard Winters being often succeeded by luxuriant Summers, and thus we find that what in Appearance causes Scarcity, may in Reality produce Plenty.

LASTLY, even Thunder, however terrible in its Appearance, and sometimes fatal in its Effects, is nevertheless very useful and beneficial upon the whole, for this likewise purifies the Air from sulphureous and oily Exhalations, and the Rains that fall with it fertilize the Earth exceedingly. It also moderates the Heat as Experience teaches us, for as it is always gloomy and sultry before Thunder, so it is afterwards generally cool and pleasant. These Remarks, though very short, may give the Reader an Opportunity of extending his Observations throughout all the Variations of Weather, and enable him to discern how useful and instructive a thing the Study of its Alterations may be, and how probable it is, that by proper Care and Attention, we may arrive at a much more useful, as well as a much more certain Knowledge in regard to the Weather, than hitherto has been attained.

FINIS.

Footnotes

a

Sol quoque & exoriens, & cum se condit in undas,
Signa dabit: *Solem* certissima signa sequuntur,
Et quæ Mane refert, & quæ surgentibus *Astris*,
Ille ubi nascentem *maculis* variaverit Ortum
Concavus in Nubem, medioque refugerit Orbe;
Suspecti tibi sint *Imbres*. Namque urget ab alto
Arboribusque satisque Notus Pecorique sinister.

Virgil. Georgic.
lib. i. v.
438.

^b Nat. Hist. lib. xviii. cap. 35.

^c Ubi supra.

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